

**Class Test 1: Rational Agents, Knowledge Base & Reasoning.****Outline**

1. Explain various aspects of the Agent concept with examples.
2. What are the characteristics of a rational agent?
3. Describe important concerns about the Environment for designing a rational agent.
4. Characterize and explain with examples Forward and Backward chaining algorithms with respect to inference in PL.
5. Give an illustrative example of limitations of logical inference.
6. Describe the syntactic rules of First Order Logic (FOL).
7. Express with an example the semantics of a query to knowledgebase in FOL.
8. How do you explain unification? Demonstrate the execution of the major steps of the simplified UNIFY algorithm using an example.
9. Explain with a suitable example the steps of conversion of a natural language sentence to a sentence in CNF of FOL.
10. State the resolution principle and demonstrate its application in proving the truth of a query to a KB in CNF of FOL.
11. Explain the resolution-refutation completeness of a KB.

1. Describe two important environment contrasts that are generally considered for designing rational agents. (2)
  2. Prove or disprove  $\exists x(\text{Mat}(x) \wedge \text{Tk}(\text{Hab}, x))$  from the following knowledgebase using resolution. (3)
    - a)  $\text{Std}(\text{Hab})$
    - b)  $\text{Mat}(\text{MatFor}(y)) \vee \neg \text{Std}(y)$
    - c)  $\neg \text{Std}(y) \vee \text{Tk}(y, \text{MatFor}(y))$
  3. Elaborate the concept of Unification? (2)
  4. Show that  
$$\forall x, y, z ((\text{Br}(x, y) \wedge \text{Sn}(z, x)) \Rightarrow \text{Nw}(z, y))$$
is a syntactically correct sentence of First Order Logic. (3)
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1. Explain the characteristics of a rational Agent. (2)
  2. Convert the following FOL sentence into CNF. (3)
$$\forall x (\text{Std}(x) \wedge \text{Tk}(x, \text{CSE1203}) \leftrightarrow \text{CStd}(x))$$
  3. Why backward chaining algorithms are said to be goal oriented? (2)
  4. Prove or disprove the proposition M using a typical forward chaining algorithm on the knowledgebase that follows. (3)
$$\begin{aligned} &A, B, C, D, G \wedge M \Rightarrow H, A \wedge H \Rightarrow E, C \wedge K \Rightarrow L, H \wedge K \Rightarrow S, B \wedge F \Rightarrow G, B \wedge D \Rightarrow K, \\ &D \wedge F \Rightarrow G, D \wedge G \Rightarrow M, D \wedge L \Rightarrow H. \end{aligned}$$
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1. How is a Term defined in the syntax of First Order Logic? (2)
  2. Resolve  $P2(\text{"Bt"}, \text{"Rt"})$  from the following clauses, considering that the variables are standardized. (3)
$$\begin{aligned} &P1(y, F1(x)) \vee P2(y, u) \\ &\neg P1(\text{"Bt"}, z) \vee P3(u, F2(x)) \\ &\neg P3(\text{"Rt"}, v) \end{aligned}$$
  3. Explain the matching rules that are involved during Unification. (2)
  4. Derive the most general unifier, if possible, for the following set of FOL sentences. (3)
$$S = \{P1(F2(\text{"Mk"}), z, \text{"Bl"}, u), P1(y, \text{"Rd"}, x, F3(z))\}$$
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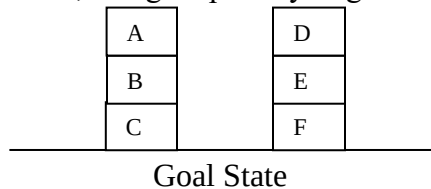
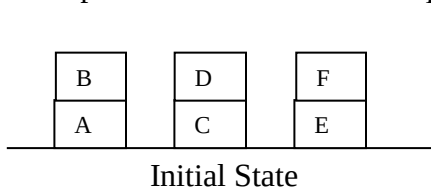
1. What are the basic components of an Agent? (2)
2. Consider the following situation of 'Find Gold avoiding Monsters' problem.
$$\begin{aligned} S_{i,j} &\Leftrightarrow M_{i-1,j} \vee M_{i+1,j} \vee M_{i,j-1} \vee M_{i,j+1} & (i \geq 1) \wedge (j \geq 1) \wedge (i \leq 4) \wedge (j \leq 4) \\ \neg M_{1,1} \wedge \neg M_{2,1} \wedge S_{2,1} \wedge \neg G_{2,1} & A_{2,1} \end{aligned}$$
Derive  $M_{3,1} \vee M_{2,2}$  using the inference rules of propositional logic. (3)
3. How Resolution is used to prove or disprove a sentence? (2)
4. Convert the following English sentence to CNF of FOL. (3)

A student of the department of EEE must take EEE1101 and PHY1201.

CSE4107  
Course Outline, Spring 2022  
Class Test 3. Planning Problems and Game Playing

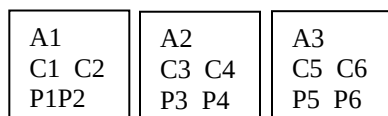
1. How are states and actions of planning problems represented using PDDL (Planning Domain Definition Language)?
2. Describe the general concepts, concerns and techniques of planning problems.
3. Illustrate with examples various aspects of problem representation and plan generation for the Air cargo Transportation and Blocks World Tower Building Problems.
4. What distinguishing features of game playing you know?
5. Describe and illustrate with an example the representation of a two-player board game.
6. Explain with an example the major steps of the MINIMAX algorithm for finding optimal game strategies.
7. Describe important aspects of the MINIMAX algorithm, and explain how such a search strategy may be improved for practical purposes.
8. Illustrate the use of Alpha-Beta pruning for efficient search strategies for real games.

1. How a planner exerts intelligence through domain feature consideration? (2)
2. Generate an optimal plan through Forward State Space search using PDDL for the planning problem in the Blocks world presented below. Omit the predicate 'Block', and give priority to gain in time over gain in actions. (3)



3. Why game playing problems are considered adversarial search problems? (2)
4. Construct a pruned game tree using Alpha-Beta pruning. Take the sequence, [6, 3, 2, 6, 4, 5, 7, 2, 8, 7, 5, 1, 3, 4] of MINIMAX values for the nodes at the cutoff depth of 4 plies. Assume that branching factor is 2, MAX makes the first move, and nodes are generated from *left to right*. (3)

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1. How 'precondition' of actions are represented with PDDL? (2)
  2. Generate an optimal plan using PDDL and Forward State Space search for the Air Cargo Transportation problem given below. Omit predicates 'Airport', 'Cargo' and 'Plane', and consider generally accepted rules and notations for such a problem. (3)



Initial State



Goal State

3. Elaborate the concept of quiescence search. (2)
4. Construct a pruned game tree using Alpha-Beta pruning. Take the sequence, [1, 3, 4, 5, 7, 4, 6, 2, 1, 3, 5, 4, 8, 4, 6] of MINIMAX values for the nodes at the cutoff depth of 4 plies. Assume that branching factor is 2, MIN makes the first move, and nodes are generated from *right to left*. (3)

## CSE4107, Spring 2022

### Chapter 3 Outline

1. Describe the basic concepts of solving AI problems as search problems.
2. What do you mean by informed or heuristic search? Cite examples of heuristic functions.
3. Explain admissibility and consistency of heuristic functions with suitable examples.
4. Describe important characteristics of greedy best-first search.
5. Illustrate with an example how priority queues and possible paths are tracked in greedy best-first search.
6. Describe the evaluation function used in A\* search.
7. Elaborate the important decisions about A\* search and also the problems with it.
8. Illustrate with an appropriate example the execution of the A\* search algorithm.
9. Explain the distinguishing features of hill-climbing local search with an example.
10. How the search environment in a hill-climbing local search is described using 'state space landscape', and what measures are taken to face the odds of the landscape?
11. What ideas stand behind genetic algorithms?
12. Describe the major steps of a typical genetic algorithm and illustrate them with examples.
13. Narrate important features of 'crossover' and 'mutation'.

## Chapter 6. Uncertainty Management and Decision Making

1. Describe in short the fundamental aspects of acting under uncertainty.
2. Explain important features and formulas of inference using full joint-probability distribution and Bayes's rule.
3. Cite illustrative examples of inference using full joint-probability distribution, Bayes's rule and joint probability of consistent models.
4. How do you explain the purpose of Bayesian Networks?
5. Explain with a concise example the specification of Bayesian Networks.
6. Explain and illustrate with an example how joint-probabilities are represented and inference is made using Bayesian Networks.
7. Narrate the basic principle of Decision making, and explain how Expected Utility of an action, given some evidence, is calculated.
8. Give an illustrative example of a decision network, and explain how utility of an action is computed and decision is made using a decision network.

Marks: 10

Time: 25 min.

1. What role does utility theory play in acting under uncertainty? (2)
2. Write a short note on Types of nodes in Decision networks. (3)
3. Consider the 'Escape the monster' environment given below.

|             |             |             |
|-------------|-------------|-------------|
| OK<br>1,2 S | OK<br>2,2 S |             |
| OK<br>1,1   | OK<br>2,1   | OK<br>3,1 S |

Assume that OK indicates absence of a monster(M) at the cell, the status of the cells of the 4 x 4 grid other than those shown are unknown, and that the agent can smell(S) a monster at a cell vertically or horizontally adjacent to it.

Find  $P(M_{2,3})$  taking 0.1 as the independent probability of a monster at any of the unknown cells. (5)

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1. What role does probability theory play in acting under uncertainty? (2)
  2. Derive the Bayes' rule from the definition of conditional probabilities, and explain its importance. (3)
  3. Say, we have a Bayesian network containing 4 random variables A, B, C and D with two distinct values each. The conditional probability tables assigned to the random variables are given below. Draw the network and compute  $P(D | a \wedge \neg b \wedge c)$  using the network. (5)

| A        | B        | P(c) |
|----------|----------|------|
| a        | b        | 0.7  |
| a        | $\neg b$ | 0.5  |
| $\neg a$ | b        | 0.2  |
| $\neg a$ | $\neg b$ | 0.02 |

| A        | C        | P(d) |
|----------|----------|------|
| a        | c        | 0.8  |
| a        | $\neg c$ | 0.4  |
| $\neg a$ | c        | 0.3  |
| $\neg a$ | $\neg c$ | 0.01 |

| B        | P(a) |
|----------|------|
| b        | 0.6  |
| $\neg b$ | 0.05 |

| P(b) |
|------|
| 0.09 |

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1. What are the major components of a Bayesian Network? (2)
  2. Explain the basic principle of decision theory. (3)
  3. Joint-probability distribution of two random variables, having three different values each is shown in the table below.

|                |                 | X <sub>2</sub>  |                 |                 |
|----------------|-----------------|-----------------|-----------------|-----------------|
|                |                 | V <sub>21</sub> | V <sub>22</sub> | V <sub>23</sub> |
| X <sub>1</sub> | V <sub>11</sub> | 0.06            | 0.07            | 0.13            |
|                | V <sub>12</sub> | 0.21            | 0.14            | 0.12            |
|                | V <sub>13</sub> | 0.08            | 0.15            | 0.04            |

Explain the major features of such a distribution regarding inference about the domain. (5)

- 
1. What are the major components of a Decision Network? (2)
  2. Explain the purpose of Bayesian Networks. (3)
  3. Joint-probability distribution of three Boolean random variables is shown in the table below.

|   |          | a     |          | $\neg a$ |          |
|---|----------|-------|----------|----------|----------|
|   |          | c     | $\neg c$ | c        | $\neg c$ |
| b | b        | 0.104 | 0.016    | 0.082    | 0.008    |
|   | $\neg b$ | 0.016 | 0.066    | 0.144    | 0.564    |

Compute the posterior probabilities,  $P(\neg c \vee b | \neg a)$  and  $P(a | \neg b \vee c)$  using the distribution. (5)

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1. What are the major tools for acting under uncertainty? (2)
  2. Explain how utility of an alternative is computed using a decision network. (3)
  3. Joint-probability distribution of two random variables, having three different values each is shown in the table below.

|                |                 | X <sub>2</sub>  |                 |                 |
|----------------|-----------------|-----------------|-----------------|-----------------|
|                |                 | V <sub>21</sub> | V <sub>22</sub> | V <sub>23</sub> |
| X <sub>1</sub> | V <sub>11</sub> | 0.06            | 0.07            | 0.13            |
|                | V <sub>12</sub> | 0.21            | 0.14            | 0.12            |
|                | V <sub>13</sub> | 0.08            | 0.15            | 0.04            |

Compute the probabilities of the compound propositions,  $V_{11} \vee V_{22}$ ,  $V_{12} | V_{23}$  and  $V_{21} | V_{13}$  using the distribution. (5)

1. What do you know about Decision nodes? (2)
2. Explain the Bayes' rule using an illustrative example. (3)
3. Say, we have a Bayesian network containing 4 random variables A, B, C and D with two distinct values each. The conditional probability tables assigned to the random variables are given below. Draw the network and compute  $P(C \mid \neg a \wedge b \wedge d)$  using the network. (5)

| A        | B        | P(d) |
|----------|----------|------|
| a        | b        | 0.7  |
| a        | $\neg b$ | 0.5  |
| $\neg a$ | b        | 0.2  |
| $\neg a$ | $\neg b$ | 0.02 |

| A        | D        | P(c) |
|----------|----------|------|
| a        | d        | 0.8  |
| a        | $\neg d$ | 0.4  |
| $\neg a$ | d        | 0.3  |
| $\neg a$ | $\neg d$ | 0.01 |

| B        | P(a) |
|----------|------|
| b        | 0.6  |
| $\neg b$ | 0.05 |

| P(b) |
|------|
| 0.09 |

1. What do you know about conditional probability tables? (2)
2. Write a brief essay on 'judgmental domains'. (3)
3. Consider the 'Escape the monster' environment given below.

|           |             |             |
|-----------|-------------|-------------|
| OK<br>1,2 | OK<br>2,2 S |             |
| OK<br>1,1 | OK<br>2,1   | OK<br>3,1 S |

Assume that OK indicates absence of a monster(M) at the cell, the status of the cells of the 4 x 4 grid other than those shown are unknown, and that the agent can smell(S) a monster at a cell vertically or horizontally adjacent to it. Find  $P(M_{3,2})$  taking 0.2 as the independent probability of a monster at any of the unknown cells. (5)

1. How probability is related to uncertainty management? (2)
2. Explain the advantages and disadvantages of inference using full joint-probability distribution. (3)
3. Say, we have a Bayesian network containing 4 random variables A, B, C and D with two distinct values each. The conditional probability tables assigned to the random variables are given below. Draw the network and compute  $P(C \mid \neg a \wedge b \wedge \neg d)$  using the network. (5)

| A        | B        | P(d) |
|----------|----------|------|
| a        | b        | 0.9  |
| a        | $\neg b$ | 0.5  |
| $\neg a$ | b        | 0.2  |
| $\neg a$ | $\neg b$ | 0.02 |

| A        | D        | P(c) |
|----------|----------|------|
| a        | d        | 0.8  |
| a        | $\neg d$ | 0.4  |
| $\neg a$ | d        | 0.3  |
| $\neg a$ | $\neg d$ | 0.01 |

| B        | P(a) |
|----------|------|
| b        | 0.6  |
| $\neg b$ | 0.05 |

| P(b) |
|------|
| 0.07 |

1. What are the positive sides of Bayesian networks? (2)
2. Illustrate the use of utility theory in Decision Networks. (3)
3. Joint-probability distribution of two random variables, having three different values each is shown in the table below.

|       |          | $X_2$    |          |          |
|-------|----------|----------|----------|----------|
|       |          | $v_{21}$ | $v_{22}$ | $v_{23}$ |
| $X_1$ | $v_{11}$ | 0.05     | 0.07     | 0.09     |
|       | $v_{12}$ | 0.22     | 0.14     | 0.13     |
|       | $v_{13}$ | 0.11     | 0.15     | 0.04     |

Compute  $P(v_{12} \vee v_{22})$ ,  $P(v_{11} | v_{23})$  and  $P(v_{21} | v_{11})$  using the distribution. (5)

1. Write a short note on Utility node. (2)
2. Describe the components of a Bayesian network. (3)
3. Joint-probability distribution of three Boolean random variables is shown in the table below.

|          | a     |          | $\neg a$ |          |
|----------|-------|----------|----------|----------|
|          | c     | $\neg c$ | c        | $\neg c$ |
| b        | 0.104 | 0.012    | 0.084    | 0.006    |
| $\neg b$ | 0.014 | 0.068    | 0.146    | 0.566    |

Compute the posterior probabilities,  $P(\neg c \vee \neg b \mid \neg a)$  and  $P(c \mid b \vee \neg a)$  using the distribution. (5)



1. How do you define a Bayesian network? (2)
2. Explain the nodes representing future state in a Decision network. (3)
3. Joint-probability distribution of two random variables, having three different values each is shown in the table below.

|                |                 | X <sub>2</sub>  |                 |                 |
|----------------|-----------------|-----------------|-----------------|-----------------|
|                |                 | V <sub>21</sub> | V <sub>22</sub> | V <sub>23</sub> |
| X <sub>1</sub> | V <sub>11</sub> | 0.05            | 0.07            | 0.09            |
|                | V <sub>12</sub> | 0.22            | 0.14            | 0.13            |
|                | V <sub>13</sub> | 0.11            | 0.15            | 0.04            |

Illustrate the features of inference using full joint-probability distribution in light of the given distribution. (5)

1. How is the probability theory related to uncertainty management? (2)
2. Explain how utility of an alternative is computed using a Decision network. (3)
3. Say, we have a Bayesian network containing 4 random variables A, B, C and D with two distinct values each. The conditional probability tables assigned to the random variables are given below. Draw the network and compute  $P(D \mid \neg a \wedge \neg b \wedge c)$  using the network. (5)

| A        | B        | P(c) |
|----------|----------|------|
| a        | b        | 0.7  |
| a        | $\neg b$ | 0.5  |
| $\neg a$ | b        | 0.2  |
| $\neg a$ | $\neg b$ | 0.02 |

| A        | C        | P(d) |
|----------|----------|------|
| a        | c        | 0.8  |
| a        | $\neg c$ | 0.4  |
| $\neg a$ | c        | 0.3  |
| $\neg a$ | $\neg c$ | 0.01 |

| B        | P(a) |
|----------|------|
| b        | 0.6  |
| $\neg b$ | 0.05 |

| P(b) |
|------|
| 0.09 |

1. How the truth of a proposition is computed involving full joint-probability distribution? (2)
2. What are the functions of Chance nodes and Decision nodes? (3)
3. Consider the 'Escape the monster' environment given below.

|           |           |           |
|-----------|-----------|-----------|
| OK<br>1,2 | S         |           |
| OK<br>1,2 | OK<br>2,2 | S         |
| OK<br>1,1 | OK<br>2,1 | OK<br>3,1 |
|           |           | S         |

Assume that OK indicates absence of a monster(M) at the cell, the status of the cells of the 4 x 4 grid other than those shown are unknown, and that the agent can smell(S) a monster at a cell vertically or horizontally adjacent to it.

Find  $P(M_{2,3})$  taking 0.3 as the independent probability of a monster at any of the unknown cells. (5)

1. Why a full joint-probability distribution can be called a complete knowledgebase of the domain. (2)
2. How utility functions are computed in Decision Networks? (3)
3. Say, we have a Bayesian network containing 4 random variables A, B, C and D with two distinct values each. The conditional probability tables assigned to the random variables are given below. Draw the network and compute  $P(C \mid a \wedge \neg d)$  using the network. (5)

| A        | B        | P(d) |
|----------|----------|------|
| a        | b        | 0.8  |
| a        | $\neg b$ | 0.5  |
| $\neg a$ | b        | 0.2  |
| $\neg a$ | $\neg b$ | 0.02 |

| A        | D        | P(c) |
|----------|----------|------|
| a        | d        | 0.9  |
| a        | $\neg d$ | 0.4  |
| $\neg a$ | d        | 0.3  |
| $\neg a$ | $\neg d$ | 0.01 |

| B        | P(a) |
|----------|------|
| b        | 0.7  |
| $\neg b$ | 0.05 |

| P(b) |
|------|
| 0.6  |

1. Write a short note on 'A degree of usefulness'. (2)
2. How do you explain the importance of Bayesian Networks? (3)
3. Joint-probability distribution of two random variables, having three different values each is shown in the table below.

|                |                 | X <sub>2</sub>  |                 |                 |
|----------------|-----------------|-----------------|-----------------|-----------------|
|                |                 | V <sub>21</sub> | V <sub>22</sub> | V <sub>23</sub> |
| X <sub>1</sub> | V <sub>11</sub> | 0.06            | 0.07            | 0.09            |
|                | V <sub>12</sub> | 0.22            | 0.14            | 0.15            |
|                | V <sub>13</sub> | 0.11            | 0.12            | 0.04            |

Compute  $P(v_{21} \vee v_{12})$ ,  $P(v_{22} \mid v_{12} \vee v_{13})$  and  $P(v_{11} \mid v_{22} \vee v_{23})$  using the distribution. (5)

1. Write a short note on 'Future state in Decision Networks'. (2)
2. How consistent models of the environment can be used for making inference in uncertain domains? (3)
3. Joint-probability distribution of three Boolean random variables is shown in the table below.

|          | a     |          | $\neg a$ |          |
|----------|-------|----------|----------|----------|
|          | c     | $\neg c$ | c        | $\neg c$ |
| b        | 0.103 | 0.015    | 0.082    | 0.007    |
| $\neg b$ | 0.016 | 0.068    | 0.144    | 0.565    |

Compute the posterior probabilities,  $P(a|c)$ ,  $P(\neg a \vee b | \neg c)$  and  $P(a | \neg b \vee c)$  using the distribution. (5)

1. How utility of an alternative is computed using a decision network? (2)
2. Explain the main problems and relevant tools for acting under uncertainty. (3)
3. Joint-probability distribution of two random variables, having three different values each is shown in the table below.

|       |          | $X_2$    |          |          |
|-------|----------|----------|----------|----------|
|       |          | $v_{21}$ | $v_{22}$ | $v_{23}$ |
| $X_1$ | $v_{11}$ | 0.06     | 0.09     | 0.13     |
|       | $v_{12}$ | 0.19     | 0.14     | 0.12     |
|       | $v_{13}$ | 0.08     | 0.15     | 0.04     |

Compute the probabilities of the compound propositions,  $v_{12} \vee v_{21}$ ,  $v_{13} | v_{23}$  and  $v_{22} | v_{12}$  using the distribution. (5)

1. What do you know about Utility functions? (2)
2. Point out the positive and negative sides of inference using full joint-probability distribution. (3)
3. Say, we have a Bayesian network containing 4 random variables A, B, C and D with two distinct values each. The conditional probability tables assigned to the random variables are given below. Draw the network and compute  $P(D | \neg b \wedge c)$  using the network. (5)

| A        | B        | $P(c)$ |
|----------|----------|--------|
| a        | b        | 0.9    |
| a        | $\neg b$ | 0.7    |
| $\neg a$ | b        | 0.2    |
| $\neg a$ | $\neg b$ | 0.02   |

| A        | C        | $P(d)$ |
|----------|----------|--------|
| a        | c        | 0.8    |
| a        | $\neg c$ | 0.4    |
| $\neg a$ | c        | 0.3    |
| $\neg a$ | $\neg c$ | 0.01   |

| B        | $P(a)$ |
|----------|--------|
| b        | 0.6    |
| $\neg b$ | 0.05   |

| $P(b)$ |
|--------|
| 0.5    |

**AHSANULLAH UNIVERSITY OF SCIENCE AND TECHNOLOGY**

Department: Computer Science and Engineering

Program: **Bachelor of Science in Computer Science and Engineering**

Semester Final Examination: Spring 2021

Year: **4<sup>th</sup>** Semester: **1<sup>st</sup>**

Course Number: **CSE4107**

Course Name: Artificial Intelligence

**Time: 3 (Three) hours**

**Full Marks: 70**

Instruction: Use **separate scripts** for **Part A** and **Part B**. Marks allotted are indicated in the right margin.

| Part A                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                    |   |   |   |   |   |   |   |   |   |   |   |       |
|-----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|---|---|---|---|---|---|---|---|---|---|---|-------|
| There are 4 (four) questions. Question #1 is compulsory. Answer any 2 (two) from other three. |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                    |   |   |   |   |   |   |   |   |   |   |   |       |
| Question 1. [Marks: 16 $\frac{2}{3}$ ]                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                    |   |   |   |   |   |   |   |   |   |   |   |       |
| a)                                                                                            | Elaborate the concept of Rational Agents.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | [3.0]              |   |   |   |   |   |   |   |   |   |   |   |       |
| b)                                                                                            | Prove or disprove the proposition X using a typical Forward Chaining algorithm on the knowledgebase that follows.<br>A, B, C, D, $G \wedge P \Rightarrow H$ , $A \wedge H \Rightarrow E$ , $H \wedge K \Rightarrow X$ , $E \wedge X \Rightarrow P$ , $C \wedge K \Rightarrow L$ , $B \wedge F \Rightarrow G$ ,<br>$B \wedge D \Rightarrow K$ , $D \wedge F \Rightarrow G$ , $D \wedge G \Rightarrow P$ , $D \wedge L \Rightarrow H$ .                                                                                                                                                            | [4.5]              |   |   |   |   |   |   |   |   |   |   |   |       |
| c)                                                                                            | Analyze the role of Hill-climbing Local Search strategy in finding reasonable solutions to hard problems with no specific goal states.                                                                                                                                                                                                                                                                                                                                                                                                                                                           | [4.5]              |   |   |   |   |   |   |   |   |   |   |   |       |
| d)                                                                                            | Present a comparative study between Greedy Best-First Search and A* Search algorithms.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | [4 $\frac{2}{3}$ ] |   |   |   |   |   |   |   |   |   |   |   |       |
| Question 2. [Marks: 15]                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                    |   |   |   |   |   |   |   |   |   |   |   |       |
| a)                                                                                            | Show using resolution that the FOL sentence $\exists x(P1(x) \wedge P2("C1", x))$ is true according to the following knowledgebase.<br>i) $\neg P3(y) \vee P2(y, F1(y))$<br>ii) $P1(F1(y)) \vee \neg P3(y)$<br>iii) $P3("C1")$                                                                                                                                                                                                                                                                                                                                                                   | [5.0]              |   |   |   |   |   |   |   |   |   |   |   |       |
| b)                                                                                            | Perform an analysis regarding use of good Heuristic functions in solving real problems with huge search spaces.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | [5.0]              |   |   |   |   |   |   |   |   |   |   |   |       |
| c)                                                                                            | Evaluate the usefulness of ideas of Evolution in Genetic algorithms for solving optimization problems.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | [5.0]              |   |   |   |   |   |   |   |   |   |   |   |       |
| Question 3. [Marks: 15]                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                    |   |   |   |   |   |   |   |   |   |   |   |       |
| a)                                                                                            | Generate an optimal plan through Forward State Space search using PDDL for the planning problem in the Blocks world presented below. Omit the predicate 'Block'.<br><div><div><table><tr><td>B</td></tr><tr><td>A</td></tr></table><table><tr><td>D</td></tr><tr><td>C</td></tr></table><table><tr><td>E</td></tr><tr><td>F</td></tr></table></div><div><table><tr><td>C</td></tr><tr><td>B</td></tr><tr><td>A</td></tr></table><table><tr><td>F</td></tr><tr><td>E</td></tr><tr><td>D</td></tr></table></div></div> <div>Initial State<span style="margin-left: 200px;">Goal State</span></div> | B                  | A | D | C | E | F | C | B | A | F | E | D | [5.0] |
| B                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                    |   |   |   |   |   |   |   |   |   |   |   |       |
| A                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                    |   |   |   |   |   |   |   |   |   |   |   |       |
| D                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                    |   |   |   |   |   |   |   |   |   |   |   |       |
| C                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                    |   |   |   |   |   |   |   |   |   |   |   |       |
| E                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                    |   |   |   |   |   |   |   |   |   |   |   |       |
| F                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                    |   |   |   |   |   |   |   |   |   |   |   |       |
| C                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                    |   |   |   |   |   |   |   |   |   |   |   |       |
| B                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                    |   |   |   |   |   |   |   |   |   |   |   |       |
| A                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                    |   |   |   |   |   |   |   |   |   |   |   |       |
| F                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                    |   |   |   |   |   |   |   |   |   |   |   |       |
| E                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                    |   |   |   |   |   |   |   |   |   |   |   |       |
| D                                                                                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                    |   |   |   |   |   |   |   |   |   |   |   |       |

|                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |   |      |   |   |     |   |          |     |          |   |     |          |          |      |   |   |      |   |   |     |   |          |     |          |   |     |          |          |      |   |      |   |     |          |      |      |      |       |
|-------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|---|------|---|---|-----|---|----------|-----|----------|---|-----|----------|----------|------|---|---|------|---|---|-----|---|----------|-----|----------|---|-----|----------|----------|------|---|------|---|-----|----------|------|------|------|-------|
| b)                      | Argue that the action schema is an efficient tool to consider specific features of diverse domains in Planners.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | [5.0] |   |      |   |   |     |   |          |     |          |   |     |          |          |      |   |   |      |   |   |     |   |          |     |          |   |     |          |          |      |   |      |   |     |          |      |      |      |       |
| c)                      | Illustrate the suitability of the MINIMAX search strategy for solving Game Playing problems.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | [5.0] |   |      |   |   |     |   |          |     |          |   |     |          |          |      |   |   |      |   |   |     |   |          |     |          |   |     |          |          |      |   |      |   |     |          |      |      |      |       |
| Question 4. [Marks: 15] |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |   |      |   |   |     |   |          |     |          |   |     |          |          |      |   |   |      |   |   |     |   |          |     |          |   |     |          |          |      |   |      |   |     |          |      |      |      |       |
| a)                      | <p>Say, we have a Bayesian network containing 4 random variables A, B, C and D with two distinct values each. The conditional probability tables assigned to the random variables are given below. Draw the network and compute <math>P(d \mid \neg a \wedge b)</math> using the network.</p> <div><table><tr><td>A</td><td>B</td><td>P(c)</td></tr><tr><td>a</td><td>b</td><td>0.9</td></tr><tr><td>a</td><td><math>\neg b</math></td><td>0.4</td></tr><tr><td><math>\neg a</math></td><td>b</td><td>0.2</td></tr><tr><td><math>\neg a</math></td><td><math>\neg b</math></td><td>0.01</td></tr></table><table><tr><td>A</td><td>C</td><td>P(d)</td></tr><tr><td>a</td><td>c</td><td>0.8</td></tr><tr><td>a</td><td><math>\neg c</math></td><td>0.5</td></tr><tr><td><math>\neg a</math></td><td>c</td><td>0.3</td></tr><tr><td><math>\neg a</math></td><td><math>\neg c</math></td><td>0.02</td></tr></table><table><tr><td>B</td><td>P(a)</td></tr><tr><td>b</td><td>0.6</td></tr><tr><td><math>\neg b</math></td><td>0.05</td></tr></table><table><tr><td>P(b)</td></tr><tr><td>0.07</td></tr></table></div> | A     | B | P(c) | a | b | 0.9 | a | $\neg b$ | 0.4 | $\neg a$ | b | 0.2 | $\neg a$ | $\neg b$ | 0.01 | A | C | P(d) | a | c | 0.8 | a | $\neg c$ | 0.5 | $\neg a$ | c | 0.3 | $\neg a$ | $\neg c$ | 0.02 | B | P(a) | b | 0.6 | $\neg b$ | 0.05 | P(b) | 0.07 | [5.0] |
| A                       | B                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | P(c)  |   |      |   |   |     |   |          |     |          |   |     |          |          |      |   |   |      |   |   |     |   |          |     |          |   |     |          |          |      |   |      |   |     |          |      |      |      |       |
| a                       | b                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 0.9   |   |      |   |   |     |   |          |     |          |   |     |          |          |      |   |   |      |   |   |     |   |          |     |          |   |     |          |          |      |   |      |   |     |          |      |      |      |       |
| a                       | $\neg b$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 0.4   |   |      |   |   |     |   |          |     |          |   |     |          |          |      |   |   |      |   |   |     |   |          |     |          |   |     |          |          |      |   |      |   |     |          |      |      |      |       |
| $\neg a$                | b                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 0.2   |   |      |   |   |     |   |          |     |          |   |     |          |          |      |   |   |      |   |   |     |   |          |     |          |   |     |          |          |      |   |      |   |     |          |      |      |      |       |
| $\neg a$                | $\neg b$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 0.01  |   |      |   |   |     |   |          |     |          |   |     |          |          |      |   |   |      |   |   |     |   |          |     |          |   |     |          |          |      |   |      |   |     |          |      |      |      |       |
| A                       | C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | P(d)  |   |      |   |   |     |   |          |     |          |   |     |          |          |      |   |   |      |   |   |     |   |          |     |          |   |     |          |          |      |   |      |   |     |          |      |      |      |       |
| a                       | c                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 0.8   |   |      |   |   |     |   |          |     |          |   |     |          |          |      |   |   |      |   |   |     |   |          |     |          |   |     |          |          |      |   |      |   |     |          |      |      |      |       |
| a                       | $\neg c$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 0.5   |   |      |   |   |     |   |          |     |          |   |     |          |          |      |   |   |      |   |   |     |   |          |     |          |   |     |          |          |      |   |      |   |     |          |      |      |      |       |
| $\neg a$                | c                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 0.3   |   |      |   |   |     |   |          |     |          |   |     |          |          |      |   |   |      |   |   |     |   |          |     |          |   |     |          |          |      |   |      |   |     |          |      |      |      |       |
| $\neg a$                | $\neg c$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 0.02  |   |      |   |   |     |   |          |     |          |   |     |          |          |      |   |   |      |   |   |     |   |          |     |          |   |     |          |          |      |   |      |   |     |          |      |      |      |       |
| B                       | P(a)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |       |   |      |   |   |     |   |          |     |          |   |     |          |          |      |   |   |      |   |   |     |   |          |     |          |   |     |          |          |      |   |      |   |     |          |      |      |      |       |
| b                       | 0.6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |       |   |      |   |   |     |   |          |     |          |   |     |          |          |      |   |   |      |   |   |     |   |          |     |          |   |     |          |          |      |   |      |   |     |          |      |      |      |       |
| $\neg b$                | 0.05                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |       |   |      |   |   |     |   |          |     |          |   |     |          |          |      |   |   |      |   |   |     |   |          |     |          |   |     |          |          |      |   |      |   |     |          |      |      |      |       |
| P(b)                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |   |      |   |   |     |   |          |     |          |   |     |          |          |      |   |   |      |   |   |     |   |          |     |          |   |     |          |          |      |   |      |   |     |          |      |      |      |       |
| 0.07                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |       |   |      |   |   |     |   |          |     |          |   |     |          |          |      |   |   |      |   |   |     |   |          |     |          |   |     |          |          |      |   |      |   |     |          |      |      |      |       |
| b)                      | Analyze the advantages and disadvantages of inference using full joint-probability distribution.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | [5.0] |   |      |   |   |     |   |          |     |          |   |     |          |          |      |   |   |      |   |   |     |   |          |     |          |   |     |          |          |      |   |      |   |     |          |      |      |      |       |
| c)                      | Explain how effectively the two components of Decision Theory are incorporated in Decision Networks.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | [5.0] |   |      |   |   |     |   |          |     |          |   |     |          |          |      |   |   |      |   |   |     |   |          |     |          |   |     |          |          |      |   |      |   |     |          |      |      |      |       |

| Part B                                             |                                                                                                                                                                   |     |        |                     |                    |       |
|----------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|--------|---------------------|--------------------|-------|
| There are 3 (three) questions. Answer any 2 (two). |                                                                                                                                                                   |     |        |                     |                    |       |
| Question 1. [Marks: 11 $\frac{2}{3}$ ]             |                                                                                                                                                                   |     |        |                     |                    |       |
| a)                                                 | Explain Oversampling and Undersampling with appropriate examples.                                                                                                 |     |        |                     | [3.0]              |       |
| b)                                                 | For a Classification task, from the Confusion Matrix, TP = 23, TN = 24, FN = 32, FP = 31. Calculate Accuracy, Precision, Recall and F Score.                      |     |        |                     | [5.0]              |       |
| c)                                                 | What are Feature Normalization and Feature Weighting? Explain how these techniques help in Classification approaches with appropriate examples.                   |     |        |                     | [3 $\frac{2}{3}$ ] |       |
| Question 2. [Marks: 11 $\frac{2}{3}$ ]             |                                                                                                                                                                   |     |        |                     |                    |       |
| a)                                                 | How does Machine Learning work?                                                                                                                                   |     |        |                     | [3.0]              |       |
| b)                                                 | Consider the following data.                                                                                                                                      |     |        |                     | [5.0]              |       |
|                                                    | Customer                                                                                                                                                          | Age | Income | No. of Credit Cards |                    | Class |
|                                                    | George                                                                                                                                                            | 35  | 35k    | 3                   |                    | Yes   |
|                                                    | Rachel                                                                                                                                                            | 22  | 50k    | 1                   |                    | No    |
|                                                    | Steve                                                                                                                                                             | 63  | 70k    | 2                   |                    | Yes   |
|                                                    | Tom                                                                                                                                                               | 59  | 40k    | 2                   |                    | No    |
|                                                    | Anne                                                                                                                                                              | 25  | 29k    | 4                   |                    | No    |
|                                                    | Predict the Class of John whose age is 37, income is 50k and no. of credit cards is 2 using the <b>K Nearest Neighbor Classification</b> approach assuming K = 3. |     |        |                     |                    |       |

| c)                                                     | What are Single Learner and Ensemble Learner approaches? Explain the significance of Ensemble Learners over Single Learners with appropriate examples.                                                                                                                                                                                                                                                                                                                                          | $[3\frac{2}{3}]$          |                               |                           |            |   |   |            |   |   |            |   |   |            |   |   |       |
|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|-------------------------------|---------------------------|------------|---|---|------------|---|---|------------|---|---|------------|---|---|-------|
| <b>Question 3. [Marks: <math>11\frac{2}{3}</math>]</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                           |                               |                           |            |   |   |            |   |   |            |   |   |            |   |   |       |
| a)                                                     | Explain how Backpropagation works in Artificial Neural Networks.                                                                                                                                                                                                                                                                                                                                                                                                                                | [3.0]                     |                               |                           |            |   |   |            |   |   |            |   |   |            |   |   |       |
| b)                                                     | <p>Consider the following data.</p> <table border="1"> <thead> <tr> <th>Object</th><th>Attribute 1 (X): weight index</th><th>Attribute 2 (Y): pH value</th></tr> </thead> <tbody> <tr> <td>Medicine A</td><td>1</td><td>1</td></tr> <tr> <td>Medicine B</td><td>2</td><td>1</td></tr> <tr> <td>Medicine C</td><td>4</td><td>3</td></tr> <tr> <td>Medicine D</td><td>5</td><td>4</td></tr> </tbody> </table> <p>Using K Means Clustering technique, partition the objects into two clusters.</p> | Object                    | Attribute 1 (X): weight index | Attribute 2 (Y): pH value | Medicine A | 1 | 1 | Medicine B | 2 | 1 | Medicine C | 4 | 3 | Medicine D | 5 | 4 | [5.0] |
| Object                                                 | Attribute 1 (X): weight index                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Attribute 2 (Y): pH value |                               |                           |            |   |   |            |   |   |            |   |   |            |   |   |       |
| Medicine A                                             | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 1                         |                               |                           |            |   |   |            |   |   |            |   |   |            |   |   |       |
| Medicine B                                             | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 1                         |                               |                           |            |   |   |            |   |   |            |   |   |            |   |   |       |
| Medicine C                                             | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 3                         |                               |                           |            |   |   |            |   |   |            |   |   |            |   |   |       |
| Medicine D                                             | 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 4                         |                               |                           |            |   |   |            |   |   |            |   |   |            |   |   |       |
| c)                                                     | Consider a document containing 100 words wherein the word <i>tiger</i> appears 3 times. Also, there are 10 million documents, and the word <i>tiger</i> appears in 1000 documents of these. Calculate the TF-IDF for the word <i>tiger</i> .                                                                                                                                                                                                                                                    | $[3\frac{2}{3}]$          |                               |                           |            |   |   |            |   |   |            |   |   |            |   |   |       |